

Multi-modal Interaction with Virtual Agents: A Pick-up and Placement Case Study

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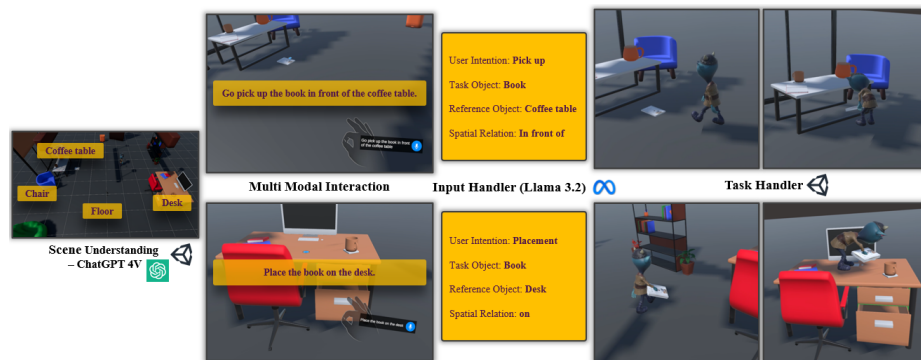


Figure 1: Framework of multi-modal interaction with a virtual agent. The user instantiates the task using multi-modal interaction. The input handler interprets the user's command and sends the information to the task handler, where, based on scene understanding, the task is executed by the agent.

ABSTRACT

This paper presents a multi-modal human-agent interaction framework that integrates hand tracking, voice commands, user intention interpretation, and scene understanding using large language and vision models for object pick-up and placement tasks in VR. Scene understanding integrates object tags and ChatGPT-4V's object detection. User commands are processed via the LLaMA-3.2 model, which interprets user intention, identifies task and reference objects, and spatial relationships for task execution. An initial experiment with 15 participants showed that the multimodal approach achieved the fastest task completion times and highest user satisfaction. While hand-only interaction was comparable, voice-only interaction was significantly slower.

Index Terms: Virtual Agent, Multi-modal Interaction, Virtual Reality, Pick-up and Placement

1 INTRODUCTION

Agents in virtual reality (VR) environments play a significant role in enhancing user interaction [1]. Research has shown that users can develop social connections with virtual characters, even when they know these characters are not real. These agents, often embodied as virtual characters, offer exciting potential across various domains [2]. For example, in educational settings, agents can serve as peers or instructors, providing interactive learning experiences, while in healthcare, they can assist in virtual rehabilitation or mental health therapies.

Interaction between virtual characters and users is central to the immersive experience in VR and to creating a sense of presence and engagement. Human interaction with the world is inherently multi-modal, involving multiple sensory modalities simultaneously. This

observation underscores the importance of developing multimodal interfaces that can closely mimic natural human behaviors to foster more intuitive and seamless user experiences [3]. Furthermore, multimodal interaction can help reduce cognitive load and improve task efficiency, as users can choose the most appropriate modality for a given task or context [4].

A single modality interaction, such as controllers or hand tracking, often struggles with effectively conveying spatial information in virtual environments. In voice interaction, it can be challenging to design a voice prompt that accurately targets a specific location, as verbal descriptions may lack the precision required for spatial tasks. In existing works, controllers are employed along with other modalities for interaction. However, the use of controllers has faced scrutiny due to several limitations [4]. These include issues of precision and fatigue caused by extended manipulation periods. To the best of our knowledge, there is a significant gap in the existing literature, with no similar studies found that explore the combination of gesture and voice as a multimodal interaction, employing large vision and language models to interpret the user's intention and ask the agent to act accordingly specifically for object pick-up and placement in virtual environments.

2 METHODOLOGY

Our approach leverages hand tracking and voice interaction to create a multimodal interaction framework for object pick-up and placement. By combining these modalities with scene understanding using vision language models and user intention interpretation by large language models, we develop a system for interacting with virtual agents in a VR environment. The process begins with scene understanding, where the system extracts the names and tags of game objects in the environment. This initial step is achieved by iterating through the game objects in the scene. However, as the agent performs pick-up and placement tasks, the scene becomes dynamic, with objects changing positions and states. The scene

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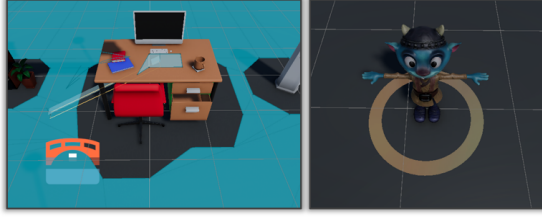


Figure 2: (left) Navigation mesh and (right) social distant preservation.

data is also sent to ChatGPT-4V¹ for object detection and verification. This dual approach allows the system to continuously update its understanding of the virtual space, covering both predefined game objects and any new spatial configurations resulting from the agent's interactions. Once the scene is understood, the user can issue a voice command that instructs the virtual agent to pick up and place an object at a desired location. The user specifies this location by gesturing with hand tracking, simulating natural human behavior. The system uses LLaMA-3.2 3B to process the voice command, extracting relevant details namely user intention (whether it is pick up or placement), task object (the object to be moved), reference object (to provide additional spatial context), and the spatial relation between them. The LLaMA model is assigned a specific role using few-shot prompt engineering to process user commands efficiently, extracting only the necessary information. After retrieving the output from LLaMA, the task handler is triggered to respond to the user's intent.

For pick-up tasks, the task handler searches for the designated task object and reference object within the scene. By applying a spatial buffer around the reference object, the handler assesses the spatial relation between the two objects. The system also verifies the user's hand-pointed location to ensure the correct object is selected for the task. Once confirmed, the handler initiates the appropriate animations and pick-up actions for the virtual agent. Similarly, for placement tasks, the handler confirms the task and reference objects, checks their spatial relationship, and ensures that the user's hand gesture accurately identifies the target location. Only after all conditions are satisfied does the agent perform the placement action, otherwise the agent sends feedback to the user.

Additionally, we implemented navigation mesh and social distance preservation (0.46m) to ensure the agent can navigate within the environment while maintaining an appropriate distance from the player, see Figure 2.

The initial evaluation of the system involved an experiment with 15 participants (9 male, 6 female, aged 21 to 29, M=25, Std=2.68) with computer education background, who were tasked with completing object pickup and placement tasks using three different interaction modalities: hand-only, voice-only, and the proposed multimodal system. We set up an indoor scene in Unity 2022.3.38² and defined four tasks containing different objects along with different spatial relations. The hand-only modality used Meta SDK V68's hand tracking to refer to objects, while the voice-only modality relied solely on voice commands. In our experiment, we utilized the Oculus Quest 2 VR headset.

3 RESULTS AND DISCUSSION

The results, shown in Table 1, demonstrate that the voice-only modality had the highest task completion time. This is likely due to the difficulty participants experienced in precisely identifying the

target location with voice commands alone, often requiring multiple attempts to complete tasks. The hand-only approach performed better, but participants still faced challenges when it came to clearly conveying spatial information using just gestures. The multimodal approach, which combined voice and gesture, resulted in the shortest task completion times. Calculating the t-test (with no correction due to the low number of comparisons), no significant difference was recognized between the hand and our approach ($P=0.053$). However, the t-test on voice and our solution showed a significant difference ($P=0.02$).

Table 1: Participants' average task completion time.

	Task Completion Time ↓			
	Task1	Task2	Task3	Task4
Hand-based	15.3	13.8	12.3	12.3
Voice-based	18.4	17.5	12.8	14.6
Ours (Multimodal)	12.1	10.7	11.8	11.7

Participants reported that the multimodal system felt more intuitive and aligned with natural human behavior, reducing the cognitive load (lower NASA-TLX score for the experimental group, 16.67) and enhancing task performance. By allowing users to simultaneously issue voice commands and point to locations with hand gestures, the system enabled more precise object pick-up and placement and improved user satisfaction.

4 FUTURE WORK

One area worth exploring is the use of the extracted text prompt from voice commands as input for generative AI models, enabling the automatic creation of virtual objects based on user commands. Additionally, further work could focus on improving the alignment and orientation of virtual objects after placement, ensuring that objects are positioned in ways that are consistent with the environment. Apart from VR settings, the exploration of such concepts in MR or robotics would be beneficial. For instance, in an industrial warehouse setting, a robotic assistant could combine voice commands, gesture recognition, and computer vision to help workers with inventory management. A worker could say "Pick up that blue box and place it on the top shelf," while pointing to the target object.

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¹<https://openai.com/index/gpt-4v-system-card/>

²<https://unity.com/>